

Naval architecture — stability, resistance & hull strength

A ship-design suite beyond the rudder / manoeuvring tools: Holtrop-Mennen resistance and powering, IMO intact and damage stability with GZ-curve area criteria, IACS hull-girder longitudinal strength and class-rule scantlings, and floating-platform stability for FPSO, semi-sub, spar, TLP and barge hulls. Every method carries its standard in the code and is pinned to published worked examples in dedicated tests.

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hull forms, live explorer

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IMO intact-stability criteria

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floating-platform types

WHAT YOU GET

- Calm-water resistance and effective power (Holtrop-Mennen) with frictional, wave-making, form and transom components, live over a 6–18 kn sweep
- GZ righting-arm curves from cross-curves ($GZ = KN - KG \cdot \sin\phi$), lost-buoyancy damage method, all six IMO intact-stability criteria — pinned to EN400 worked conditions
- IACS UR S11 vertical wave bending moment, permissible bending stress 175/k, and the S11A partial-factor ultimate check
- Prescriptive plate-thickness and lateral-pressure scantling checks to class rules
- Seakeeping natural periods, simplified RAOs and spectral RMS response; intact / damaged platform stability with wind-heel and area criteria